Course enrollment tracker

Workflow steps

step 1: data ingestion

- load cleaned dataset from:
    - delta table (/delta/course_tracker)
    - or csv (/filestore/course_tracker_csv)
- data includes:
    - student name
    - course name
    - enrollment date
    - completion percentage

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step 2: data processing

- check for valid completion percentage
- ensure values are between 0 and 100
- handle missing values

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step 3: filtering low progress students

- condition:
- completion_percentage < 50
- output:
    - list of students requiring attention

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step 4: report generation

- generate a csv report:
    - filename: weekly_low_progress_report.csv
- columns:
    - student name
    - course name
    - completion percentage

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step 5: logging

- log pipeline execution:
    - start time
    - end time
    - number of students flagged
- example log:
- pipeline started at 10:00 am
- total students flagged: 3
- pipeline completed successfully

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python script

```
import pandas as pd

df = pd.read_csv("course_tracker_csv")

low_progress = df[df['completion_percentage'] < 50]

low_progress.to_csv("weekly_low_progress_report.csv", index=False)

print("pipeline executed successfully")

print("students flagged:", len(low_progress))
```

Simulated azure devops yaml pipeline

```
trigger:
- weekly
pool:
  vmImage: 'ubuntu-latest'
steps:
- script: echo "starting pipeline"
  displayName: "start"
- script: python report_script.py
  displayName: "generate report"
- script: echo "pipeline completed"
  displayName: "end"
```